

ANAESTHETIC DRUGS

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OVERVIEW

- ▶ Anaesthesia is the loss of all sensation while analgesia is the selective loss of pain sensation
- ▶ Local anaesthetic drugs block the conduction of nerve impulses in the peripheral nerves or spinal cord
- ▶ Local anaesthetics are used to anaesthetize a particular part or region of the body and have proved useful in patients undergoing surgery on the skin and subcutaneous tissue, eyes, ears, joints
- ▶ General anaesthetic drugs block the cortical neuronal activity responsible for consciousness and all sensations
- ▶ General anaesthetic drugs therefore, prevent consciousness and cause amnesia during major surgical procedures
- ▶ The presence of amnesia prevents patients from recalling the surgical procedure

LOCAL ANAESTHETIC (LA) DRUGS

3

ESTER TYPE DRUGS

- ▶ Cocaine
- ▶ Benzocaine
- ▶ Chloroprocaine

AMIDE TYPE DRUGS

- ▶ Lidocaine
- ▶ Bupivacaine
- ▶ Etidocaine
- ▶ Prilocaine
- ▶ Ropivacaine

- Based on the chemical structure, LA drugs are divided into either ester or amide types
- Because LA drugs act directly at the site of administration, their duration of action is dependent on the rate of diffusion and absorption from the site of administration
- In some formulations, epinephrine is added to prolong the duration of action for LA drugs by producing vasoconstriction which subsequently reduces its rate of absorption
- However, because of risk of ischaemia and necrosis, LA drugs with epinephrine are not used in tissues with end arteries like the fingers, toes, ears, nose and penis

Mechanism of action for LA drugs

- ▶ LA drugs cause a reversible inhibition of action potential conduction by binding to the sodium channels and decreasing the nerve membrane permeability to sodium
- ▶ Each LA drug has a lipophyllic (hydrophobic) portion and a hydrophilic portion
- ▶ The non polar (lipophilic) form of the anaesthetic drug molecule passes through the neuronal membrane and switches to the polar (hydrophilic) form in the cytoplasm of the neuron
- ▶ This cationic form of the anaesthetic binds to the cytoplasmic side of the sodium channel protein thereby, prolonging the inactivation state of the sodium channel
- ▶ With sodium channels blocked, action potentials cannot propagate along the neuronal fibre and sensory input is lost

Adverse effects of LA Drugs

Adverse effects mostly come about due to the systemic absorption of local anesthetic drugs

5

- ▶ CNS stimulation; restlessness, tremor and euphoria
- ▶ CNS inhibition; drowsiness and sedation
- ▶ Cardiovascular depression; hypotension
- ▶ Headache, nausea, parasthesias
- ▶ Seizures and subsequent comas are experienced in high doses
- ▶ Hypersensitivity reactions especially the ester type which are usually metabolized to para amino benzoic acid(PABA) which causes allergic reactions

NB: Most local anaesthetic drugs have anti-arrhythmic activity but toxic levels suppresses cardiac conduction leading to tachy arrhythmias

Indications of LA Drugs

- ▶ It should be stressed that LA drugs can either be administered parenterally or sometimes applied topically, the route mostly being dependent on the site of anaesthesia

Topical anaesthesia

- ▶ Used to anaesthetize the skin, mucous membrane or cornea
- ▶ Treatment of pruritus, caused by poison ivy, insect bites, eczema
- ▶ Relieving of pain caused by oral, laryngeal or rectal disorder or surgery

Infiltration anaesthesia

- ▶ The most common route involving the injecting of anaesthetic drug direct into subcutaneous tissue just under the skin
- ▶ Useful for minor procedures e.g. suturing of wound, removal of foreign bodies and or dental procedures
- ▶ When administering LA as an infiltration, epinephrine can be added but as earlier mentioned, it should be avoided tissues with end arteries

Iontophoresis

- ▶ This technique uses a small electric current to force molecules of LA into the tissue
- ▶ Dentists have found it to be very useful and is also approved in children

Nerve block and field anaesthesia

- ▶ These are forms of regional anaesthesia which is achieved by blocking sensory nerve conductivity of that area by injecting the drug into or adjacent to a peripheral nerve or nerve plexus
- ▶ Examples of nerves blocked are the radial nerve involving the portions of the forearm and the hand, orbital nerve for the ocular surgery
- ▶ In field anaesthesia, the LA drug is administered in a series of injections to form a wall of anaesthesia encircling the operative field

Spinal intrathecal anaesthesia

- ▶ Used to block somatosensory and motor fibers during surgical procedures on the lower limb or pelvic structure
- ▶ The LA drug is injected in the subarachnoid, intrathecal space below the level at which the spinal cord terminates
- ▶ Due to possible cerebral spinal leakage from the lumbar puncture, spinal anesthesia has been reported to cause headaches
- ▶ In cases where the LA drug ascends too high the spinal cord, respiratory depression has been reported

Epidural Anaesthesia

- ▶ This is produced by injecting a LA drug into the lumbar or caudal epidural (Extradural) space
- ▶ A LA drug like bupivacaine is usually administered through this route to provide anaesthesia during labour and delivery
- ▶ After epidural administration the LA is absorbed into the systemic circulation and hence doses must be carefully monitored to avoid cardiac depression and neurotoxicity in the mother and neonate

SPECIFIC AGENTS

10

Ester – Type LA (Cocaine, procaine, Tetracaine, Benzocaine)

- ▶ Cocaine was the first natural discovered LA drug while procaine was the first synthetic LA drug
- ▶ Has both LA and CNS stimulating effects with the most significant vasoconstrictive effect
- ▶ Because of CNS effects and potential for abuse, cocaine is seldomly used as LA but is often used to anaesthetize internal structure of the nose where its vasoconstrictive effects helps prevent bleeding after nasal surgery
- ▶ Cocaine and procaine are both metabolized to PABA and thus are more likely to cause allergic reaction
- ▶ Tetracaine has longer duration of action than procaine and is used in infiltration and is also available as a topical spray
- ▶ Benzocaine is a frequently used topical anesthetic and is available for treatment of sunburn, pruritus and other skin conditions

Amide –Type LA (Lidocaine, prilocaine, bupivacaine, mepivacaine)

11

- ▶ Lidocaine is the most widely used LA drug and is available in a number of formulations such as topical solutions and ointments, oral sprays, viscous gels for oral or laryngeal use and parenteral preparations
- ▶ Lidocaine is also used as infiltration, nerve block, epidural and spinal anaesthesia
- ▶ A eutectic mixture of lidocaine and prilocaine is available as a cream to anaesthetize intact skin to a depth of 5mm while in paediatric patients, it is used before venipuncture, I.V cannulation or circumcision
- ▶ Bupivacaine, mepivacaine and ropivacaine all have similar clinical use but differ in their duration of action with bupivacaine being the most widely used in obstetric anaesthesia and also the cardiac depression occurring more frequently than with the other agents
- ▶ Bupivacaine is also available as a liposomal encapsulated formulation for long acting analgesia useful in management of postsurgical pain

GENERAL ANAESTHETIC DRUGS

INTRODUCTION

13

- ▶ General anesthetics depress the central nervous system to a sufficient degree to permit the performance of surgery and other noxious or unpleasant procedures
- ▶ Not surprisingly, general anesthetics have low therapeutic indices and thus require great care in administration
- ▶ While all general anesthetics produce a relatively similar anesthetic state, they are quite different in their secondary actions (side effects) on other organ systems

CLASSIFICATION OF GENERAL ANAESTHETICS

14

They are classified into two on the basis of their route of administration i.e. inhalation and intravenous anesthetics

1. INHALATION ANAESTHETICS:

- Rapidly absorbed from lungs alveoli
- Cross blood-brain-barrier
- Offer minute to minute control over the depth of anaesthesia

They include:

i. Gaseous Anaesthetics e.g. Nitrous Oxide

ii. Volatile liquid anaesthetics:

- Halothane, Methoxyflurane, Isoflurane, Desflurane, Sevoflurane
- Diethyl ether and Vinyl ether

2. INTRAVENOUS ANAESTHETICS:

- ▶ Intravenous anesthetics act much more rapidly, producing unconsciousness in about 20 seconds, as soon as the drug reaches the brain from the site of its injection.

- ▶ These agents used for induction of anaesthesia followed by inhalation agent.

- ▶ The main induction agents in current use are:
 - I. Ultra-short acting barbiturates e.g. Thiopentone
 - II. Short acting benzodiazepines e.g. Midazolam, Diazepam
 - III. Opiate analgesics e.g. Morphine, Fentanyl
 - IV. Others e.g. Ketamine and Propofol

CLASSIFICATION OF GENERAL ANAESTHETIC

Inhalational Anaesthetics

Nitrous oxide
Halothane
Isoflurane
Desflurane
Sevoflurane

Parenteral Anaesthetics

Thiopental
Midazolam
Ketamine
Propofol
Etomidate
Fentanyl

Other agents

Dantrolene

Selection of the route of administration

- ▶ The selection of specific drugs and routes of administration to produce general anesthesia is based on;
 - I. pharmacokinetic properties
 - II. The secondary effects in the context of the proposed diagnostic or surgical procedure with the consideration of the individual patient's age, associated medical condition, and medication use

General principals on surgical anaesthesia

18

- ▶ Unlike the practice of every other branch of medicine, anaesthesia is usually neither therapeutic nor diagnostic though status asthmaticus can be treated with halothane while intractable angina can be treated with epidural local anesthetics

- ▶ Administration of general anaesthesia and discovery of new ones are driven by the following three general objectives;
 - I. Minimizing the potentially deleterious direct and indirect effects of anesthetic agents and techniques

 - II. Sustaining physiologic homeostasis during surgical procedures that may involve major blood loss, tissue ischemia, reperfusion of ischemic tissue, fluid shifts, exposure to a cold environment, and impaired coagulation

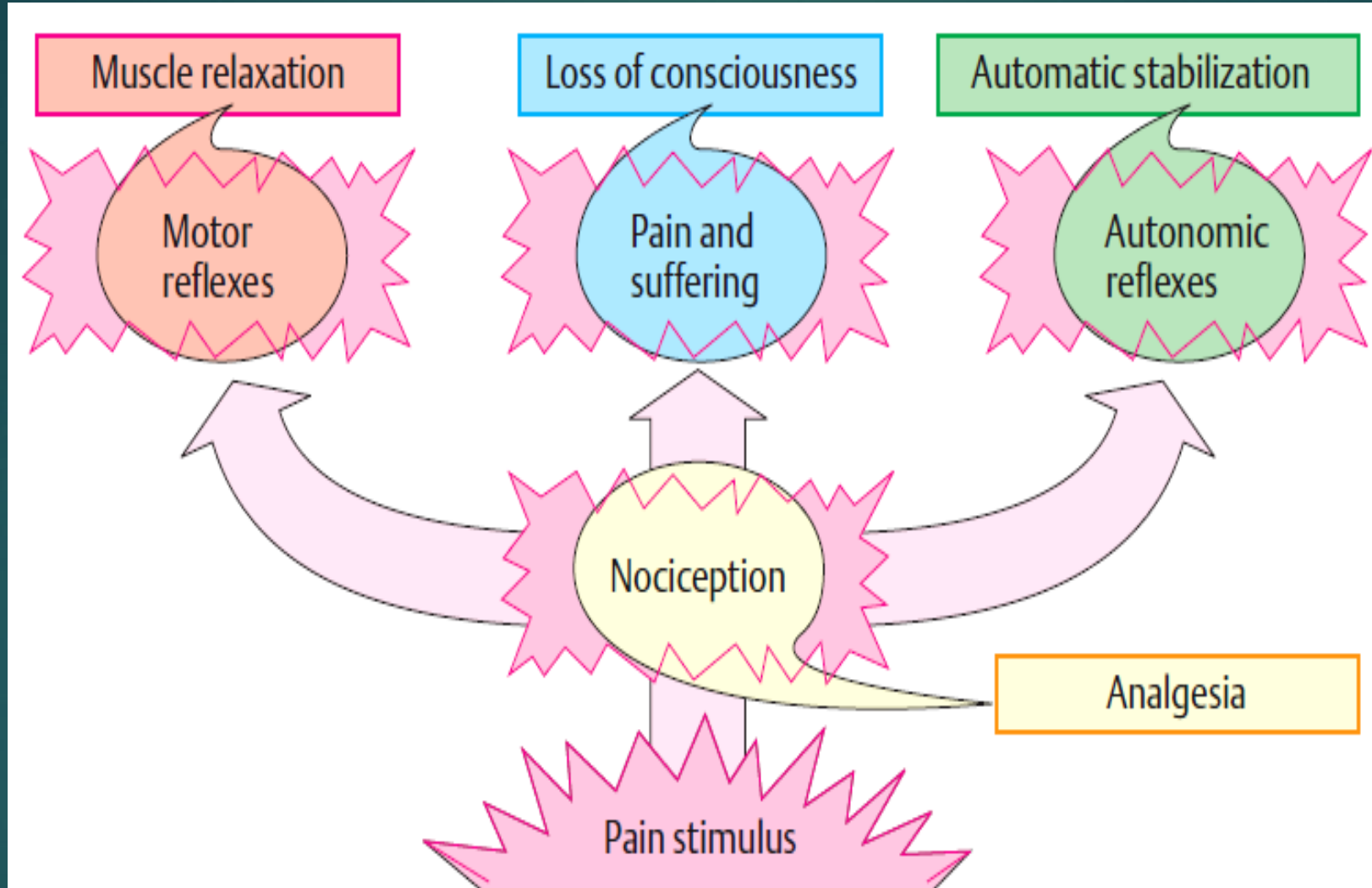
 - III. Improving postoperative outcomes by choosing techniques that block or treat components of the surgical stress response

AIM OF GENERAL ANAESTHESIA

Suppression of CNS functions while maintaining the function of other vital body organs

REQUIREMENTS FOR GENERAL ANAESTHESIA

1. Loss of consciousness: analgesia, skeletal muscle relaxation
2. Suppression of visceral reflexes with maintenance of physiological stability



GENERAL EFFECTS OF GENERAL ANAESTHESIA

21

Hemodynamic Effects of General Anesthesia

- ▶ Decrease in systemic arterial blood pressure is the most prominent physiological effect of anesthesia induction associated with the majority of both intravenous and inhalational agents
- ▶ This is thought to be due to the following;
 - Direct vasodilation
 - Myocardial depression
 - Blunting of baroreceptor control
 - Generalized decrease in central sympathetic tone

Respiratory Effects of General Anesthesia

22

- ▶ Airway maintenance is essential following induction of anesthesia, as nearly all general anesthetics reduce or eliminate both ventilatory drive and reflexes that maintain airway patency
- ▶ Therefore, ventilation must generally be assisted or controlled for at least some period during surgery
- ▶ Thus, muscle relaxation is valuable during the induction of general anesthesia where it facilitates management of the airway, including endotracheal intubation
- ▶ Neuromuscular blocking agents are commonly used to effect such relaxation, reducing the risk of coughing or gagging during laryngoscopic-assisted instrumentation of the airway, and thus reducing the risk of aspiration prior to secure placement of an endotracheal tube

- ▶ Patients commonly develop hypothermia (body temperature $<36^{\circ}\text{C}$) during surgery which could be due to low ambient temperature, exposed body cavities, cold intravenous fluids, altered thermoregulatory control, and reduced metabolic rate
- ▶ General anesthetics lower the core temperature set point at which thermoregulatory vasoconstriction is activated to defend against heat loss through their vasodilative effects which offsets cold-induced peripheral vasoconstriction, thereby redistributing heat from central to peripheral body compartments, leading to a decline in core temperature
- ▶ Prevention of hypothermia has emerged as a major goal of anesthetic care with methods which include using warm intravenous fluids, heat exchangers in the anesthesia circuit, forced-warm-air covers

Nausea and Vomiting

- ▶ Nausea and vomiting in the postoperative period are caused by an action of anesthetics on the chemoreceptor trigger zone and the brainstem vomiting center, which are modulated by serotonin (5-HT), histamine, acetylcholine, and dopamine
- ▶ The 5-HT₃-receptor antagonist ondansetron is very effective in suppressing nausea and vomiting and other common treatments also include droperidol, metoclopramide, dexamethasone, and avoidance of N₂O
- ▶ The use of propofol as an induction agent and the nonsteroidal anti-inflammatory drug ketorolac as a substitute for opioids may decrease the incidence and severity of postoperative nausea and vomiting.

MEASUREMENT OF GENERAL ANAESTHESIA

25

- ▶ General anesthetics should produce an immobilized patient and their potency is measured by determining the concentration of general anesthetic that prevents movement in response to surgical stimulation
- ▶ For inhalational anesthetics, potency is measured in MAC units (minimum alveolar concentration) that prevents movement in response to surgical stimulation in 50% of subjects
- ▶ The potency of intravenous anesthetic agents is somewhat more difficult to measure because there is no available method to measure blood or plasma anesthetic concentration continuously
- ▶ Generally, the potency of intravenous agents is defined as the free plasma concentration (at equilibrium) that produces loss of response to surgical incision (or other end points) in 50% of subjects

Inhalational anaesthetics

26

1. Halothane:

- Is the most widely used agent, highly lipid soluble, potent
- It causes arrhythmia, hangover and the risk of liver damage is high if used repeatedly

2. Nitrous oxide: - Odourless and colourless gas

- It is rapid in action and also an effective analgesic agent
- Its potency is low, hence must be combined with other agents
- It is a relatively free of serious unwanted effects

3. Enflurane: Halogenated ether (similar to halothane)

- Poorly metabolized in the liver and thus, less toxic than halothane
- It is faster in its action, less liable to accumulate in the body fat compared to halothane
- It causes seizure during induction and following recovery from anaesthesia

1. **Thiopentone: barbiturate with very high lipid solubility**

- Drug enters to tissues with a large blood flow (liver, kidneys, brain, etc) and more slowly to muscle
- Uptake into body fat occurs slowly because of the low blood flow to this tissue, which may cause prolonged effect if given repeatedly
- It causes cardiovascular depression

2. **Etomidate:**

- It is more quickly metabolized and the risk of cardiovascular depression is less compared to thiopentone
- Etomidate suppresses the adrenal cortex, which has been associated with an increase in mortality in severely ill patients

3. Ketamine:

- Acts more slowly than thiopentone and produces a different effect, known as dissociative anaesthesia in which there is a marked sensory loss and analgesia, as well as amnesia and paralysis of movement, without actual loss of consciousness
- Ketamine causes dysphoria and hallucinations during recovery

4. **Benzodiazepines** e.g. diazepam, lorazepam, midazolam

- Compared with intravenous barbiturates, benzodiazepines produce a slower onset of central nervous system effects
- Benzodiazepines prolong the post anesthetic recovery period but also cause a high incidence of amnesia for events occurring after the drug is administered
- The benzodiazepines are useful in anesthesia as premedication and intraoperative sedation

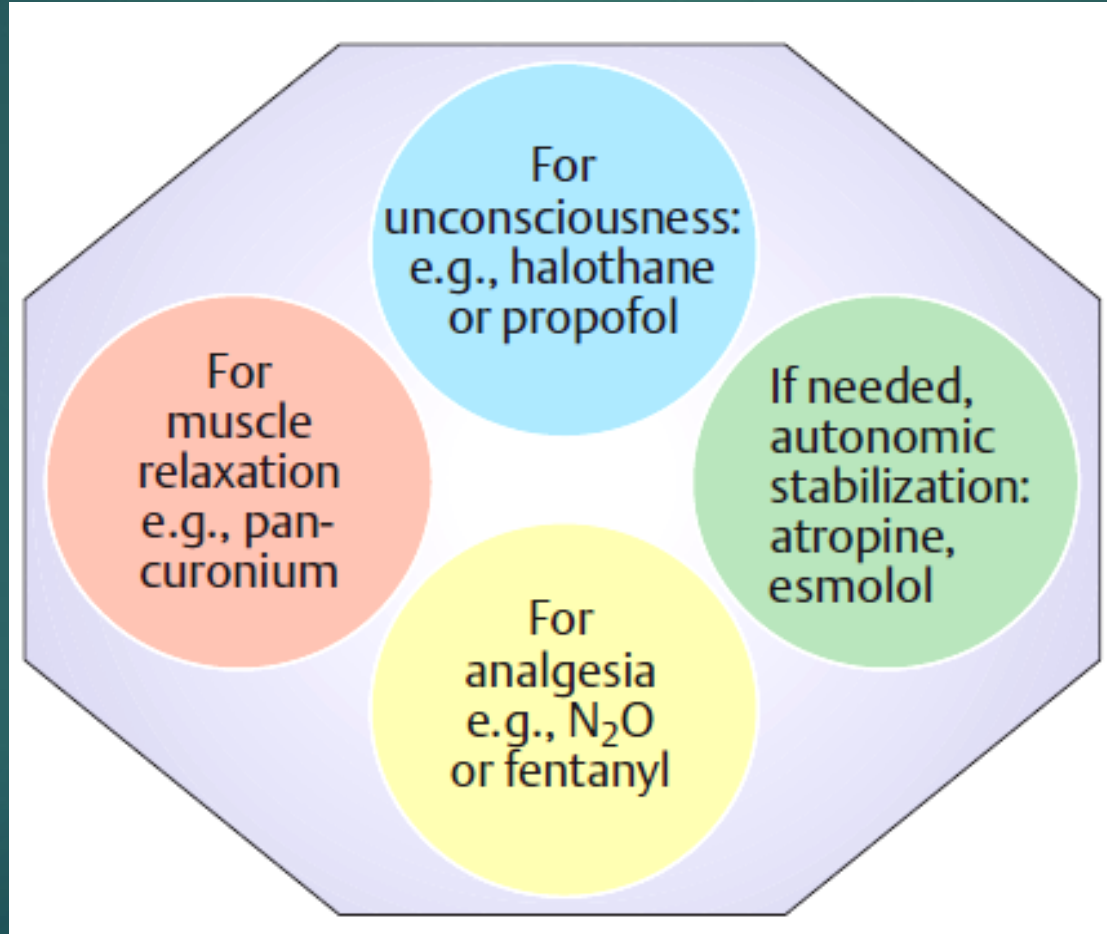
5. **Opioid analgesic anesthesia:**

- Opioid analgesics can be used for general anesthesia, in patients undergoing cardiac surgery and fentanyl and its derivatives are commonly used for these purposes

Balanced anaesthesia

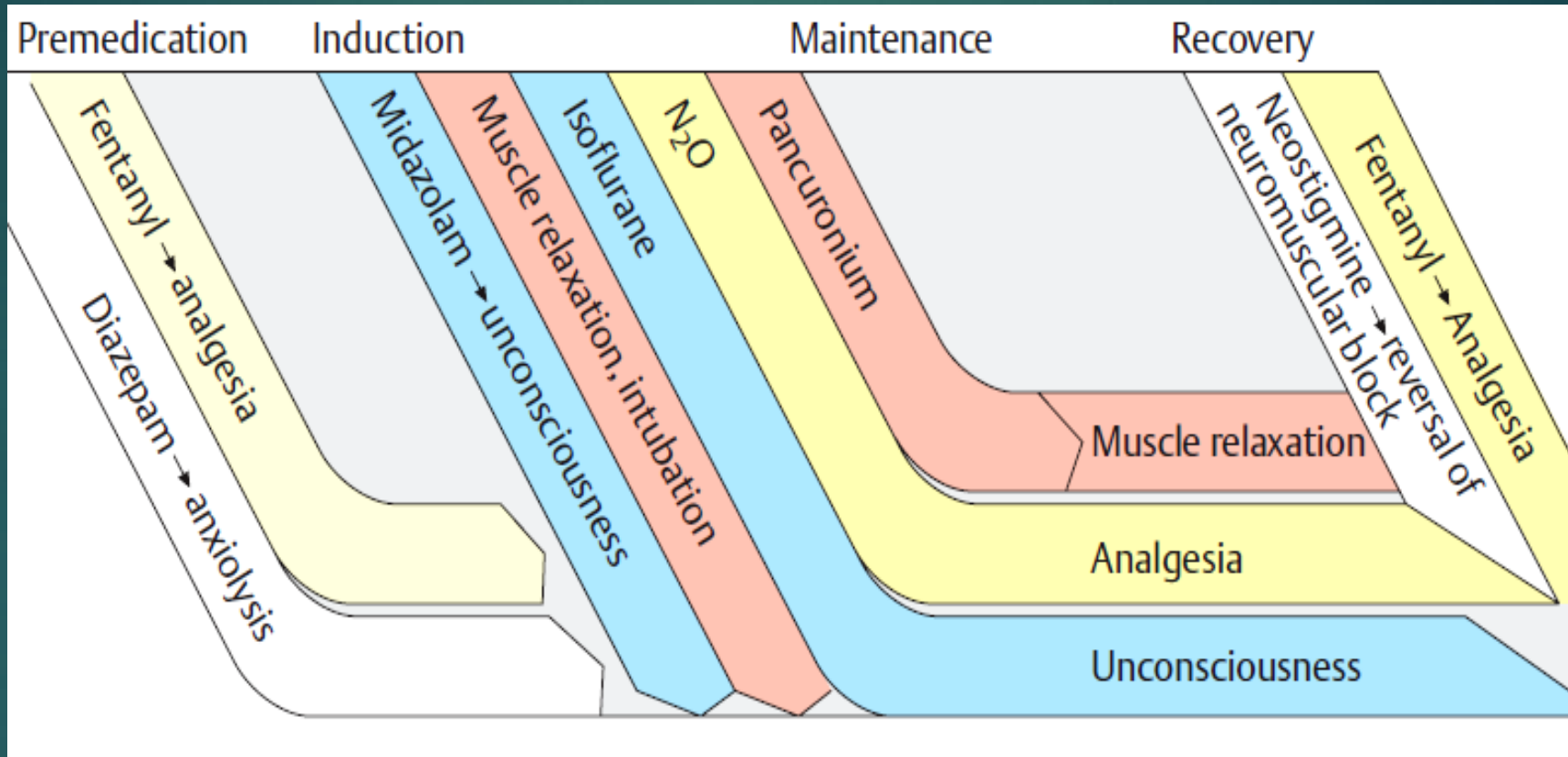
(Combination therapy)

30



Regimen for Balanced Anaesthesia

31



Adverse Effects of General Anaesthetics

- ▶ Respiratory depression
- ▶ Bronchospasm & lung atelectasis
- ▶ Hypotension, Bradycardia, Disrhythmias
- ▶ Nephrotoxicity
- ▶ Halothane-induced hepatitis
- ▶ Uterine relaxation with uterine bleeding
- ▶ Hypersensitivity reactions

THE END